GENAI BASICS • LESSON 06 • MODULE 2: MULTIMODALITY (MODULE FINALE)

Speech & Audio AI

Voice Synthesis, Whisper, Foley Sound Design & AI Music

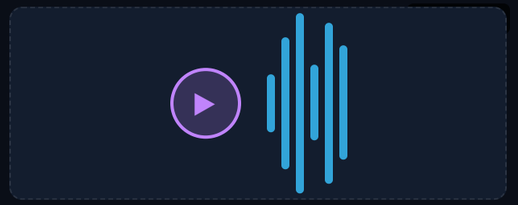
From 1D acoustic pressure waves and mel-spectrograms to discrete neural codecs (EnCodec/DAC), word-level timestamps in Whisper, ElevenLabs and Suno production consoles, Google's audio stack, and synchronized multi-stem audio with Sidechain Ducking.

THE 10/90 AXIOM IN GENERATIVE AUDIO: Generating raw acoustic waveforms with a neural network is only **10% of production success**. The remaining **90%** is Voice Activity Detection (VAD), LUFS loudness compliance, word-level subtitle alignment, anti-spoofing security, sub-150ms conversational latency, and multi-stem audio engineering.

INTERACTIVE HOOK • BLIND LISTENING TEST

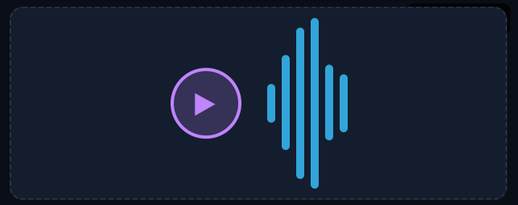
The Auditory Turing Test: Can Your Ears Detect 2026 AI Audio?

Four professional audio recordings. Write in the chat: which clips are physical microphone recordings, and which were synthesized by neural networks?



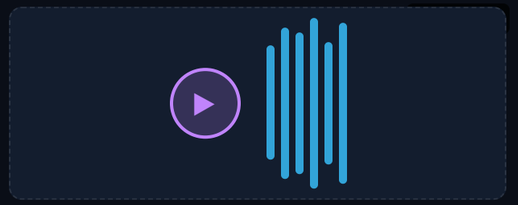
Studio Voiceover

Rich documentary voice, subtle breath intakes, natural dental sibilants.



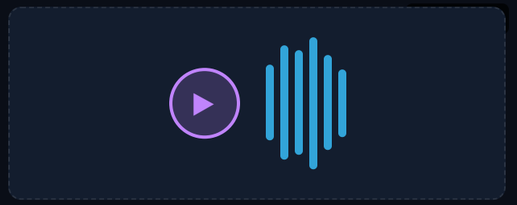
Bicycle Foley SFX

Tires splashing violently through rain puddle, chain rattle, wet pavement.



Acoustic Jazz Trio

Grand piano, brushed snare drum, acoustic walking upright bass.



Support Phone Call

130ms conversational turn-taking, conversational barge-in, natural filler words.

> **Chat Question:** Which of these 4 clips are 100% AI generation? Give students 45 seconds to submit answers.

EVOLUTION OF NEURAL AUDIO • 2020–2026

All 4 Clips Are 100% AI! The Journey Beyond Robotic Comb-Filtering

In just six years, speech and audio AI advanced from metallic GPS navigators to human-grade conversational voice agents.

2020–2021 • ROBOTIC BUZZ

WaveNet & FastSpeech

Snapshot: Harsh metallic comb-filtering, flat robotic cadences, 16kHz bandwidth limitation.

2022–2023 • ASR & STUDIO PROSODY

OpenAI Whisper & ElevenLabs v1

Snapshot: Whisper solves speech recognition at scale; ElevenLabs introduces emotional expressiveness and voice cloning.

2024–2025 • NEURAL CODECS & MUSIC

Suno v3, Udio & EnCodec

Snapshot: Full 3-minute stereo arrangements with vocals; text-to-Foley effects (SFX); discrete acoustic tokens.

2026 • ON-DEVICE VOICE AGENTS

Cartesia Sonic, Whisper v3 & Suno v4

Snapshot: Sub-150ms conversational turnaround, studio 48kHz fidelity, instantaneous speech translation preserving original timbre.

> **The 2026 Frontier:** Timbre and naturalness are solved problems. The industry battle has shifted to **ultra-low latency (<150ms)** and **deterministic multi-stem mixing**.

COURSE PROGRESSION • MODULE 2 FINALE

Concluding Multimodality: Bringing Mute Clips from L05 to Life

Module 2 connects text prompts to the physical sensory world. L04 captured frames, L05 synthesized motion. Today we complete the sensory experience.

M1 • COMPLETE
LLM Core
L01–L03: Prompts & APIs

L04 • COMPLETE
Images
C2_rain, packshot assets

L05 • COMPLETE
Video
I2V, cameras, cuts

L06 • HERE
Audio & Speech
TTS, Whisper, Foley, Music

M3 • NEXT
Ecosystem
L07: Copilot, M365 & Dev

M4–M6
Agents & Code
L08–L16: MCP, Production

WHAT WE INHERITED FROM LESSON 05

Silent 5-Second Video Takes

Our Glovo courier in the rain (`shot2_bike_rain.mp4`) and rotating coffee mug (`packshot_360.mp4`) look cinematic but remain completely mute.

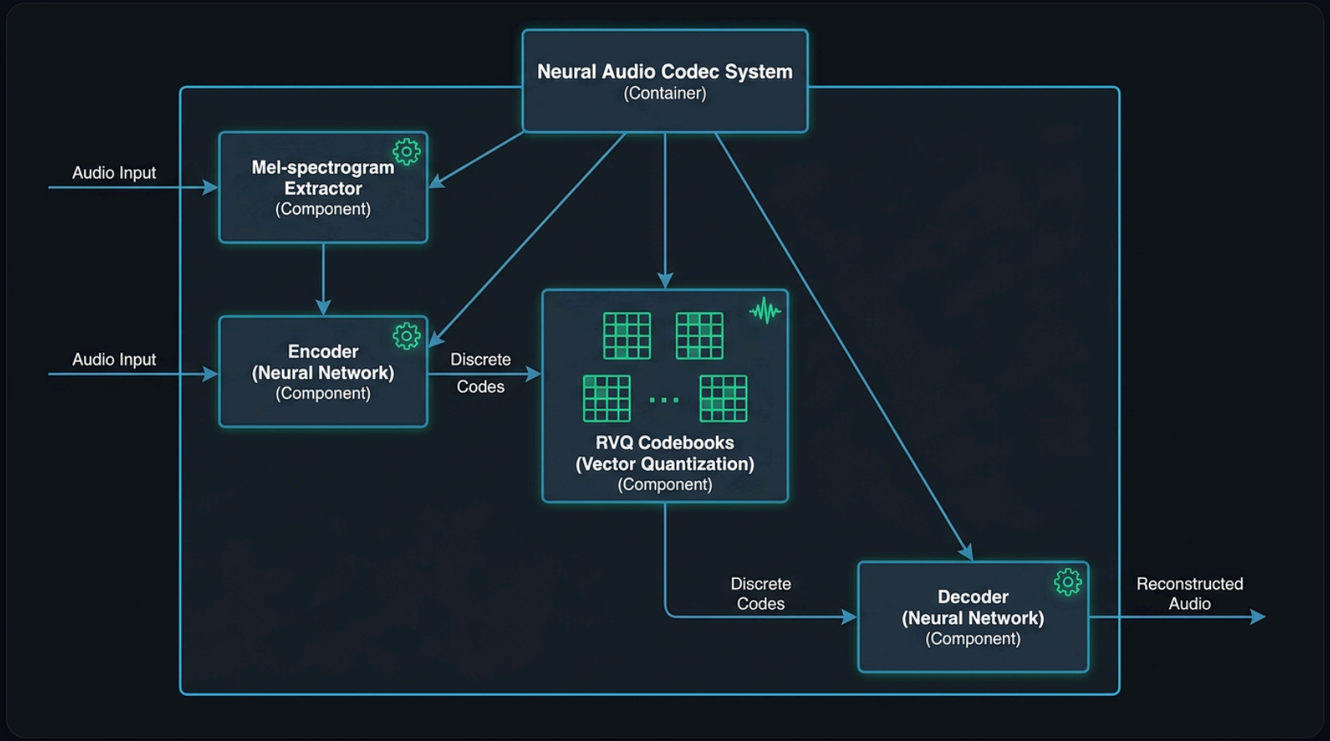
WHAT L06 DELIVERS FOR FULL PRODUCTION

3-Stem Synchronized Master Soundtrack

1. **Voiceover Narration:** Clear, studio-articulated narrative lines.
2. **Foley SFX:** Splashing water, bicycle hub rattle, ambient rain reflections.
3. **Background Music:** Thematic score with automated sidechain ducking.

Sound Physics: Waveforms, Mel-Spectrograms, and Neural Audio Codecs

How modern Transformer architectures model continuous 48kHz audio via Residual Vector Quantization (RVQ).



1. CONTINUOUS 1D PCM WAVEFORM

44,100 Float Samples / Sec

Sound is a continuous air-pressure wave. Feeding raw sample floats directly into an LLM context window would exhaust memory in two seconds.

2. SHORT-TIME FOURIER TRANSFORM (STFT)

2D Mel-Spectrograms

Decomposes audio into overlapping 25ms windows mapped onto psychoacoustic frequency bins. This serves as Whisper's input representation.

3. NEURAL CODECS (ENCODEC / DAC)

RVQ: Down to ~50 Tokens / Sec

Residual Vector Quantization: Compresses 48kHz studio audio through hierarchical codebooks into discrete tokens that standard Transformer decoders can predict autoregressively!

OpenAI Whisper: Dual-Tower Transformer and Word-Level Subtitle Alignment

The global standard for speech recognition: from 30-second audio windows to frame-accurate subtitle timing.

AUDIO ENCODER (CONV + ATTENTION)

30-Second Audio Chunks

Consumes 80/128-channel mel-spectrograms. Two 1D convolutional layers downsample time 4x, followed by self-attention blocks extracting acoustic embeddings.

TEXT DECODER (PREFIX TOKENS)

Task Control Prefix Prompts

Autoregressive decoder steered by prefix tokens: `<|startoftranscript|>`, language `<|en|>`, task `<|transcribe|>` or `<|translate|>` (direct English translation).

WORD-LEVEL TIMESTAMPS

Subtitle Automation

Flag `word_timestamps=True` analyzes Cross-Attention weight matrices using Dynamic Time Warping (DTW) to compute exact millisecond boundaries for every word.

Production Impact: Word-level timestamps enable automated Python pipelines to generate burn-in video subtitles and kinetic text animations without manual keyframing.

Whisper Playground Console: Eliminating Silence Loops via Silero VAD

Why raw Whisper hallucinates during silence and how Silero VAD paired with CTranslate2 solves it.



FAILURE: INFINITE REPETITION LOOPS

Hallucinations During Pauses

When speech stops, Whisper's autoregressive decoder latches onto background hiss and enters infinite hallucination loops: "*Thank you for watching! Like & subscribe...*".

MITIGATION: SILERO VAD (1MB NET)

Upstream Silence Stripping

Voice Activity Detection: Scans audio in 2ms, strips silence, and sends verified voice chunks to Whisper. Completely prevents repetition loops!

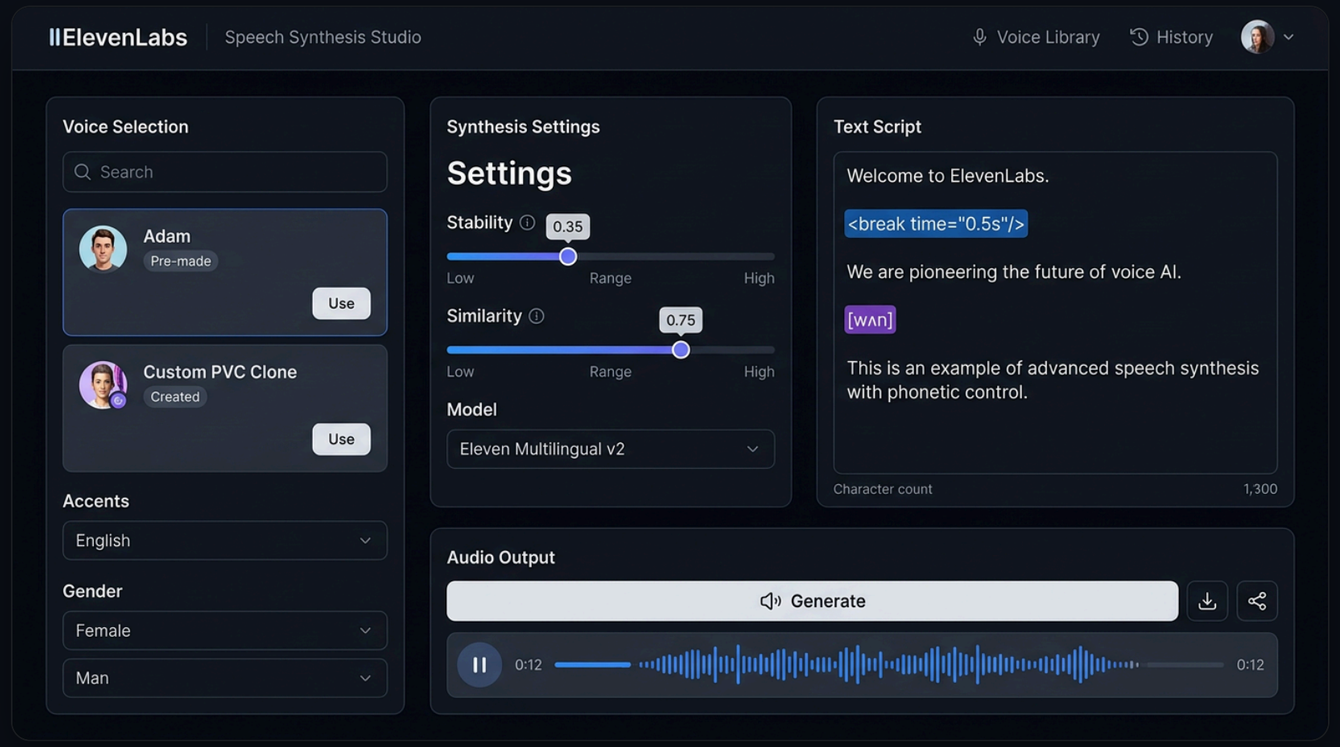
RUNTIME: FASTER-WHISPER (CTRANSLATE2)

4x Speedup via C++ & INT8

INT8/FP16 quantization reduces VRAM by 50% while accelerating transcription throughput 4x over vanilla PyTorch.

TTS Landscape 2026: Studio Warmth vs. Sub-150ms Conversational Latency

The speech synthesis market is divided into high-fidelity studio models, ultra-low latency conversational agents, and local open-weight engines.



1. STUDIO CLOUD BENCHMARK (ELEVENLABS V2)

48kHz Studio Warmth, 32+ Languages

Benchmark for video narration. Studio UI: voice selector (Adam, Custom PVC Clone), Stability & Similarity sliders, text editor with SSML phonetic tags. Latency ~350ms.

2. REAL-TIME AGENTS (CARTESIA SONIC)

135ms Time-to-First-Audio

State-Space Model architecture engineered specifically for telephony agents and live conversational systems.

3. OPEN-WEIGHT LOCAL (KOKORO-82M)

82M Params, Apache 2.0, $0 API Cost

Runs locally on CPU or in-browser via WebAssembly with zero data exfiltration.

Directing Cadence: Punctuation as Musical Notation, SSML, and Stability

Controlling narrator breath intake, cadence pauses, and emotional intensity without phonetic degradation.

PUNCTUATION AS MUSICAL NOTATION

Breaths, Pauses & Emphasis

- **Ellipses (`...`):** Forces the acoustic model to take an audible breath and decelerate pacing.
- **Em-Dashes (`—`):** Introduces a dramatic cadence pause before key narrative reveals.
- **CAPITAL LETTERS:** Signals pitch accentuation and prosodic emphasis.
- **SSML & IPA Tags:** `<break time="0.5s"/>` and exact phonetic tags: `<phoneme alphabet="ipa" ph="ɡloʊvoʊ">Glovo</phoneme>`.

ELEVENLABS SLIDER CALIBRATION

Stability: Dramatic Range vs. Monotone

- **Stability = 0.30–0.40:** High emotional range, dynamic whispers, and trailer-style theatrical acting.
- **Stability = 0.80–0.90:** Uniform, authoritative delivery suited for legal disclaimers and IVR menus.
- **Similarity Boost = 0.75:** High speaker fidelity without amplifying reference microphone noise floor.

> **Production Rule:** Never set Stability to 1.0 (results in robotic monotone) or below 0.20 (causes random pitch cracking). Commercial video sweet spot: **0.35–0.45**.

Voice Cloning: Instant (d-Vectors) vs. Professional (LoRA PVC)

Two fundamentally different approaches: fast zero-shot speaker conditioning vs. deep weight fine-tuning.

INSTANT VOICE CLONING (IVC / ZERO-SHOT)

60-Second Audio Sample • d-Vector

- **Mechanism:** Extracts a 512-dimensional speaker embedding vector and conditions synthesis on the fly.
- **Speed & Cost:** Instant setup (5s), $0 fine-tuning overhead.
- **Pitfall:** ~80% acoustic similarity. Room reverberation and laptop microphone hiss become permanently baked into every generation!

PROFESSIONAL VOICE CLONING (PVC / FINE-TUNING)

45 Minutes Studio Recording • LoRA Weights

- **Mechanism:** Fine-tunes model weights (LoRA) over 2–4 hours on enterprise GPUs using dry studio stems.
- **Outcome:** 98% fidelity, ability to whisper, shout, and reproduce intimate vocal formants.
- **Governance:** Requires active identity verification via Voice Captcha.

SECURITY & ETHICS • BIOMETRIC DEFENSE

Audio Security: Threat Vectors, Dynamic Voice Captcha, and SynthID

Protecting enterprise systems from voice fraud and unauthorized voice cloning.

2026 THREAT VECTORS

Fraud & Deepfakes

• **CEO Voice Fraud:** Vishing targeting corporate finance for urgent wire transfers.
• **Biometric Spoofing:** Defeating banking voice authentication.
• **Fabricated Evidence:** Synthetic compromising audio of public figures.

DYNAMIC VOICE CAPTCHA

Consent Verification

Platform generates a dynamic, randomized cryptographic phrase with the current timestamp that the voice owner must read live into camera.

IN-BAND WATERMARKING

Google SynthID & PerTh

Imperceptible phase modulations embedded into harmonics. Undetectable by human ears, yet mathematically verifiable even after MP3 compression and street noise re-recording!

Blast Radius in Audio: Green, Yellow, and Red Risk Boundaries

Clear boundaries for deploying neural audio across enterprise applications.

GREEN ZONE (AUTONOMOUS)

Foley, Drafts & Consented Dubbing

- Text-to-Foley environmental sound effects.
- Scratch voiceovers for internal prototype reels.
- Creator-authorized video dubbing (Creator Tools case).

Damage Risk: 0%. Fully Autonomous.

YELLOW ZONE (HUMAN-IN-THE-LOOP)

Customer Bots & Brand Ads

- Frontline support voice bots (mandatory disclosure: "*Speaking with an AI assistant*").
- Broadcast brand commercials (mandatory audio QA sign-off by sound lead).

Reputational Risk: Moderate. Human-in-the-Loop.

RED ZONE (STRICTLY FORBIDDEN)

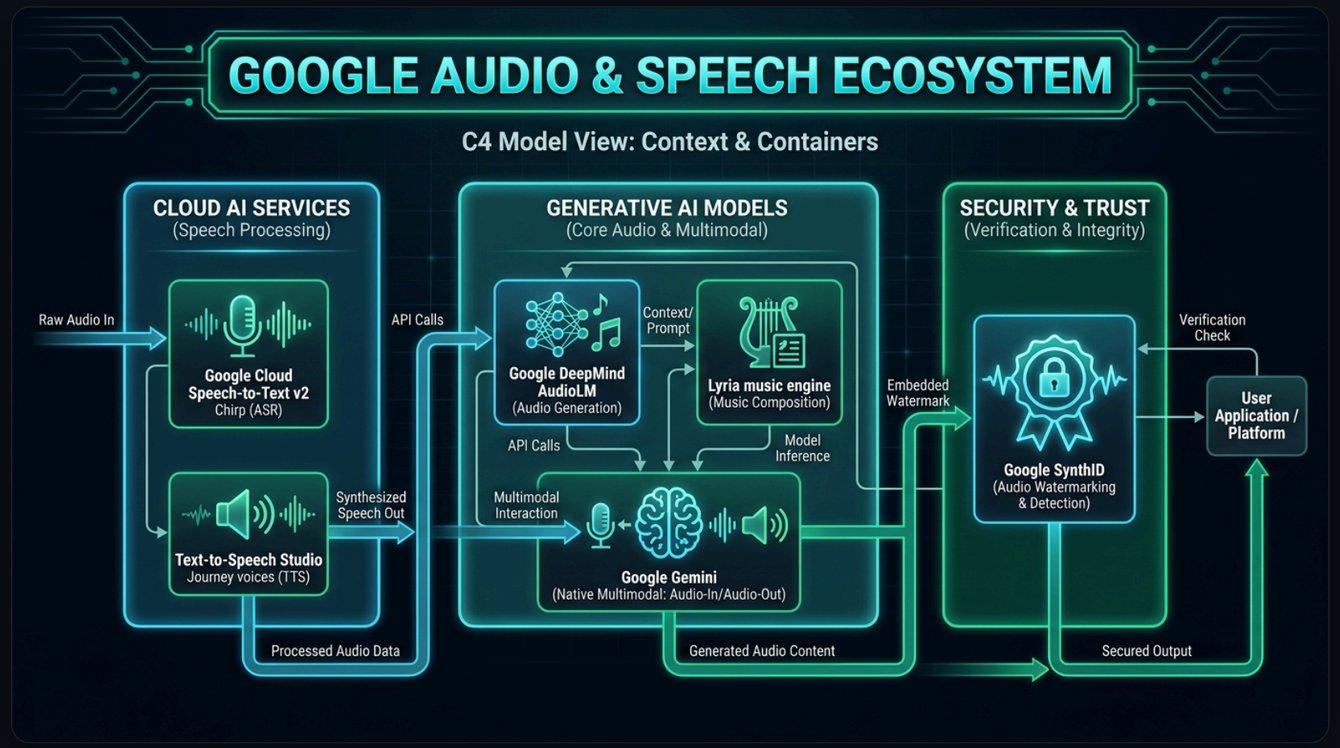
Unconsented Cloning & Spoofing

- Cloning living persons without explicit legal consent.
- Emergency services impersonation.
- Bypassing financial voice biometrics.

Strictly Forbidden. Immediate legal liability.

Google's Audio & Speech Ecosystem: Cloud Studio, DeepMind, and Gemini Native Audio

From enterprise Cloud Speech APIs to DeepMind's foundation audio models and native multimodal audio in Gemini.



1. GOOGLE CLOUD SPEECH STUDIO

Journey Voices & Chirp (USM 2B)

TTS: Journey and Studio Voices — 48kHz studio-trained neural models with SSML tuning.
ASR: Chirp model (Universal Speech Model 2B) provides speech recognition across 100+ languages in high-noise environments.

2. DEEPMIND AUDIO & LYRIA MUSIC

SoundStream, AudioLM & Lyria

DeepMind pioneered **SoundStream** (RVQ neural codecs) and **Lyria** — their state-of-the-art music engine powering YouTube Dream Track and MusicFX.

3. GEMINI NATIVE MULTIMODAL AUDIO

Direct Audio-In ➔ Audio-Out

Cascade-free intelligence: Gemini perceives raw audio tokens directly without intermediate text conversion — hearing sarcasm, accents, laughter, and street noise! Includes **SynthID** watermarking.

HANDS-ON AI R&D CASE STUDY • CREATOR TOOLS

Speech-to-Speech: Multilingual Dubbing Across 100+ Languages with Timbre Lock

The engineering pipeline behind automated video localization: LLM syllable budgeting and phase vocoders.

STAGE 1: TIMING EXTRACTION

Whisper + Silero VAD

Extracts frame-accurate word boundaries and eliminates pauses to prevent timeline drift.

STAGE 2: SYLLABLE BUDGET

LLM Syllable Budgeting

Enforces strict syllable density limits during translation to prevent timeline expansion in target languages.

STAGE 3: TIMBRE PRESERVATION

Voice Conversion

Synthesizes target dialogue while preserving original speaker formants, pitch contours, and breath dynamics.

STAGE 4: SCENE LOCK

Elastic Time-Stretch

Phase Vocoder performs ±15% tempo adjustments without pitch distortion to match on-screen video cut points.

The Outcome: Viewers in Germany, Brazil, or Japan hear the creator speaking their native language with frame-accurate video synchronization!

MIDPOINT INTERMISSION • 5 MINUTES

Auditory Sensory Reset

Processing acoustic architectures is cognitively demanding. Mute your audio, drink water, and give your ears complete silence.

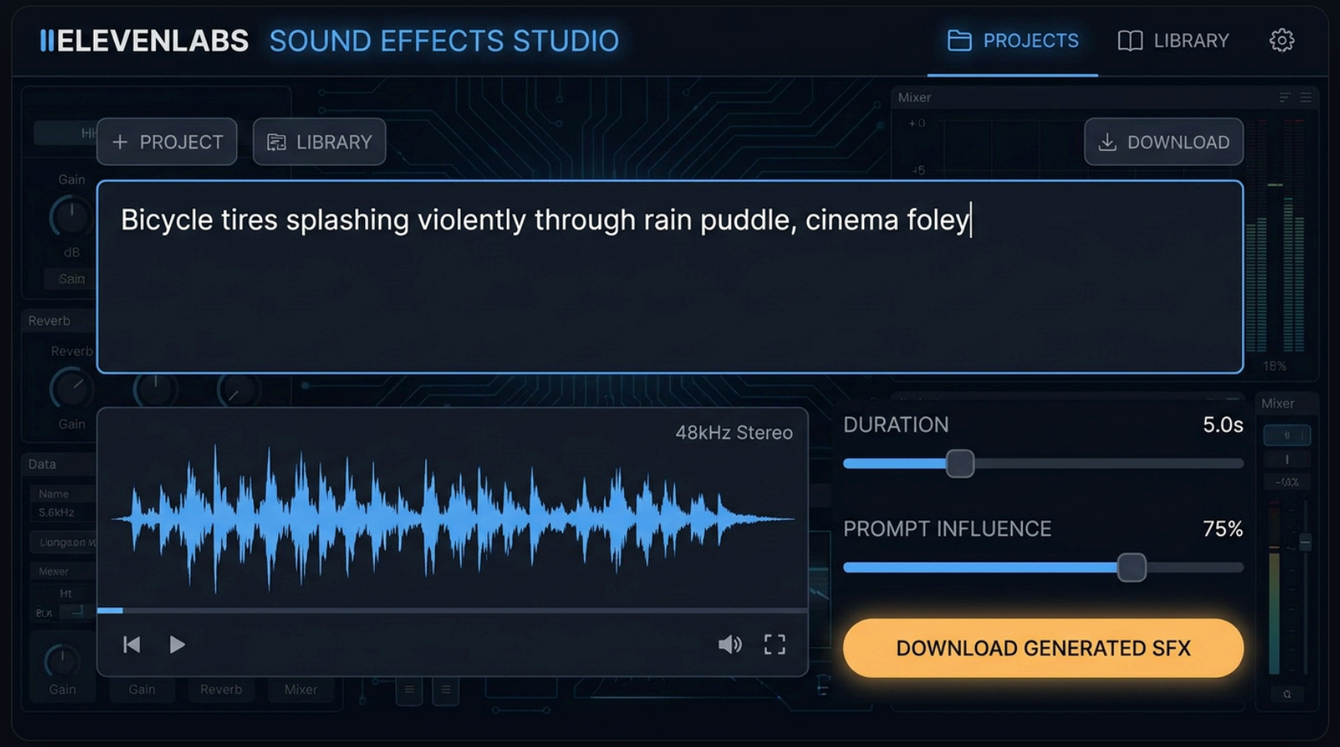
05:00

COMING UP IN PART TWO:

Foley SFX generation (ElevenLabs SFX) • Giant Stories storyboard sound matrix • AI music in Suno v4 • 3-stem mixing with Sidechain Ducking in Python • Sub-300ms real-time voice agents.

Physical Sound Design: Generating Foley from Text via the 3-Slot Formula

Eliminate hours of stock library hunting. Generative SFX models synthesize physical acoustic textures directly from material descriptions.



THE 3-SLOT FOLEY PROMPT FORMULA

[Source Material] + [Kinetic Action] + [Acoustic Space]

1. **Material:** `rubber tires`, `ceramic mug`, `wet cobblestone`.
2. **Kinetic Motion:** `splashing violently`, `scraping stone`.
3. **Space Acoustics:** `outdoor alleyway reflections`, `dry soundstage`.

BENCHMARK PROMPT FOR GLOVO COURIER

Road Bike Splash Through Rain Puddle

```
Bicycle rubber tires splashing violently through a deep rainwater puddle on wet cobblestone, fine water spray, quiet evening city background hum, cinema foley
```

Production Sound Matrix: Creator Tools Giant Stories Case Study

How real-world studios map Foley sound layers and musical hooks to each 5-second video cut before generation.

SHOT #	VISUAL SCENE FOCUS	VOICEOVER LINE	FOLEY SFX LAYER	MUSIC UNDERSCORE HOOK
Shot 1	Wax-sealed wooden crate from 1929	"Untouched for almost a century..."	Hollow stone vault resonance, dripping water, rat skitter	Low bowed contrabass drone, 60 BPM, minor
Shot 2	Oil derrick drilling in Texas heat	"Digging blind, risking everything..."	Cicadas buzzing, heavy shovel scraping dry rock	Pulsing sub-bass beat every 2s, sparse percussion
Shot 3	Rotary newspaper printing press	"Until the morning headlines broke..."	Mechanical gear rhythm, ink spray hissing, paper flutter	Rising tremolo strings, accelerating tempo
Shot 4	Glovo courier speeding in rain	"Hot dinner delivered, whatever the storm."	Violent water spray, rain slicker rustle, bicycle hub click	Warm lo-fi chillhop beat, 74 BPM, ducked under speech

> **Cinematic Engineering Axiom:** Sound design is planned **before final editing**. Each shot possesses an acoustic specification across dialogue, Foley, and musical accents.

AI MUSIC GENERATION • SUNO V4 STUDIO CONSOLE

Soundtrack Engineering: Instrumental Toggle, Arrangement Tags, and Demucs

Synthesizing controlled background music without frequency collisions against dialogue.



MANDATORY RULE: INSTRUMENTAL TOGGLE ON

No Vocals Under Narration!

AI vocal lines clash directly with voiceover speech formants, completely destroying dialogue intelligibility. Always force instrumental mode!

STRUCTURAL ARRANGEMENT TAGS IN LYRICS

Steering Compositional Dynamics

Even in instrumental mode, the lyrics window directs arrangement: `[Intro: Chillhop acoustic]` → `[Verse: 74 BPM]` → `[Build-up: Cello]` → `[Outro: Fade out]`.

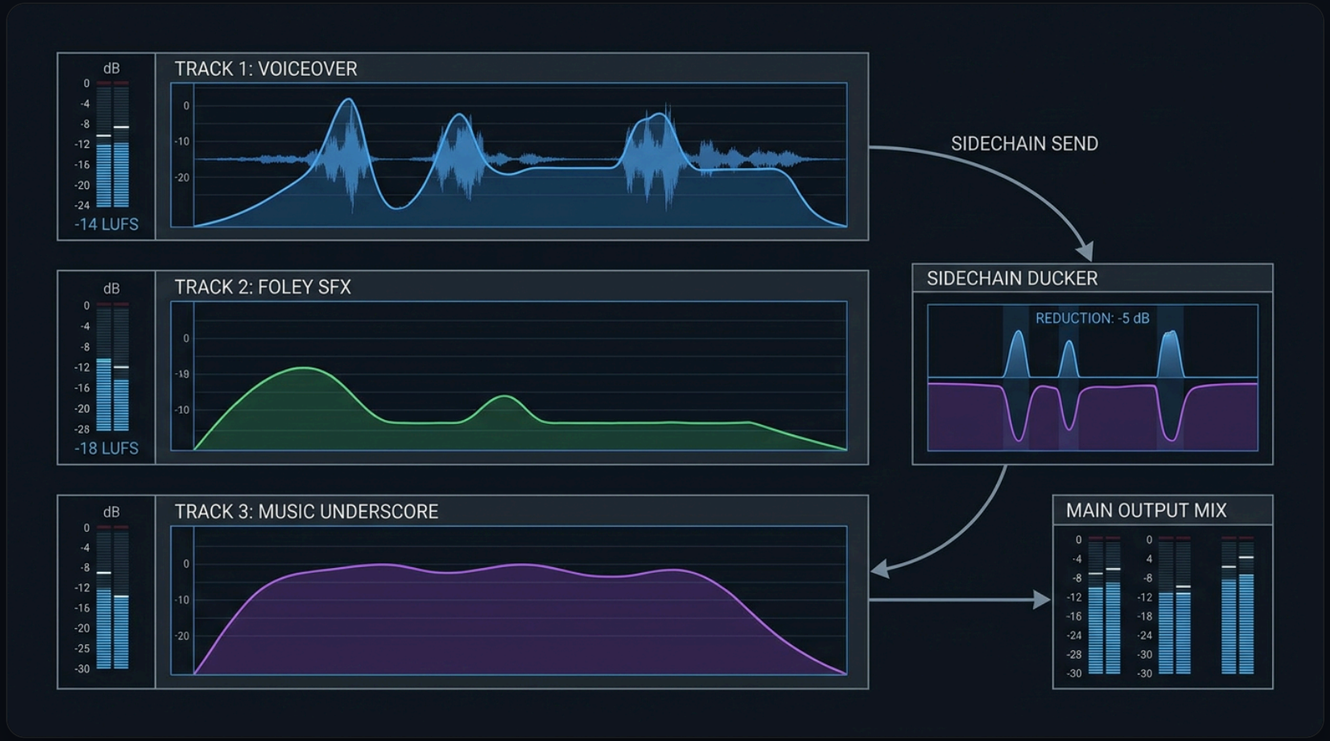
STEM SEPARATION (DEMUCS V4)

Isolate Stems in One Click

Splits generated audio into 4 stems: drums, bass, instruments, and vocals for independent fader control and ducking.

3-Stem Mixing & Sidechain Ducking: Automated Music Attenuation

Blending voiceover narration, Foley sound effects, and background music into a broadcast master without distortion.



STEM 1 • VOICEOVER NARRATION

-14 LUFS (True Peak < -1.0 dBFS)

Center mono placement. High-pass filter below 80Hz to eliminate microphone rumble.

STEM 2 • FOLEY SFX

-18 LUFS (Stereo Field Placement)

Synchronized to video cuts: water splash transient matches bicycle wheel hitting the puddle.

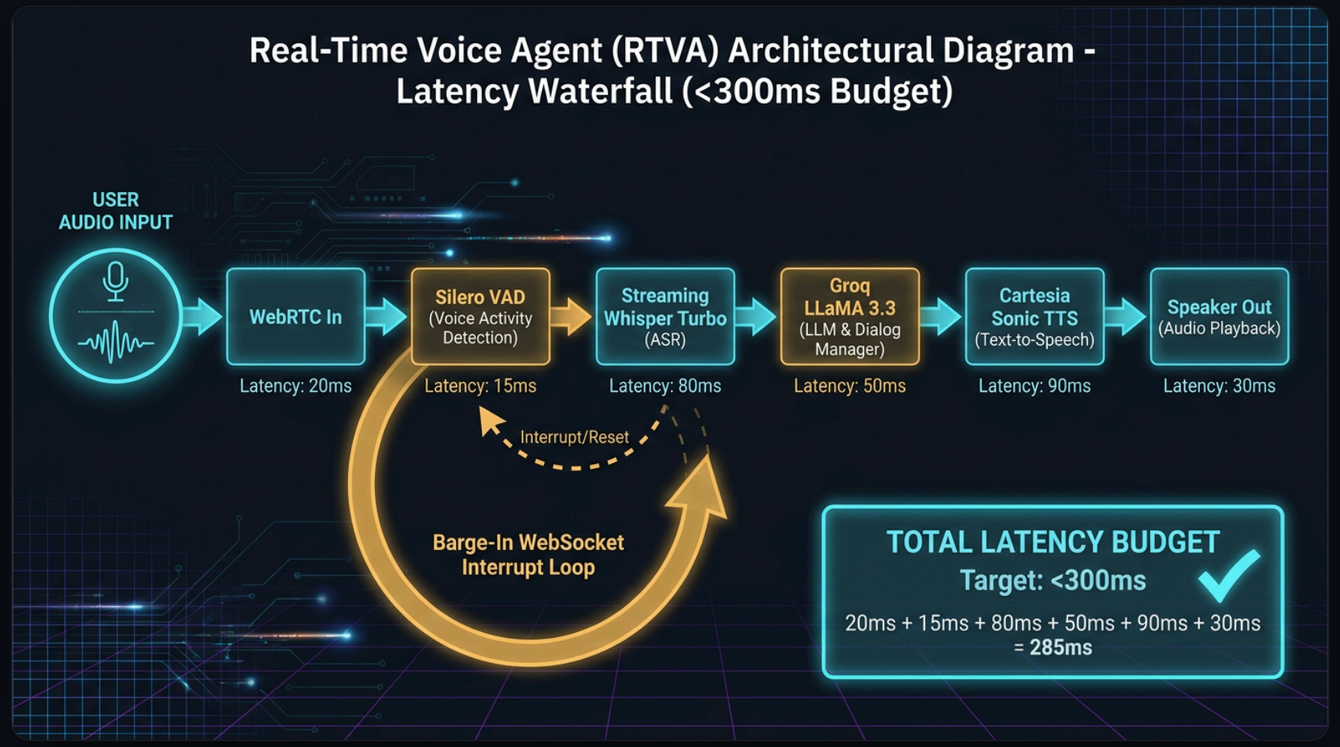
STEM 3 • MUSIC WITH SIDECHAIN DUCKING

Dynamic -5 dB Gain Reduction

Music sits at -18 LUFS during pauses. When narration speaks, automation ducks music down to **-23 LUFS**. Implemented in Python via `pydub`.

Real-Time Voice Agents: Sub-300ms Latency Budget and Barge-In Handling

Why standard batch APIs fail for conversational agents and how streaming WebSockets achieve sub-300ms turnaround.



LATENCY BUDGET WATERFALL: 285 MS

Meeting the Human Perceptual Threshold

1. WebRTC Audio Ingest: **20ms**
2. Silero VAD End-of-Speech: **15ms**
3. Streaming Whisper Turbo STT: **80ms**
4. Groq LLaMA 3.3 First Token: **50ms**
5. Cartesia Sonic Streaming TTS: **90ms**
6. Client Playback Buffer: **30ms**

BARGE-IN INTERRUPT CIRCUIT

Instant Audio Cancellation

When the user speaks over the bot, client VAD fires an interrupt signal over WebSocket. The server cancels inference, and speaker playback terminates within 20ms.

Code Automation: Whisper Word Timestamps and Multi-Stem Overlay

Production script from our Colab notebook: from transcription parsing to synchronized video export.

BLOCK 1: WHISPER WORD-LEVEL TIMESTAMPS

```
import whisper

model = whisper.load_model("large-v3-turbo")
result = model.transcribe("voice.wav", word_timestamps=True)

for segment in result["segments"]:
    for word in segment["words"]:
        print(f"[{word['start']:.2f}s -> {word['end']:.2f}s] {word['word']}")
```

BLOCK 2: MULTI-STEM OVERLAY IN PYDUB

```
from pydub import AudioSegment

voice = AudioSegment.from_file("voice.wav") - 2  # -14 LUFS
sfx   = AudioSegment.from_file("sfx.wav") - 6    # -18 LUFS
music = AudioSegment.from_file("music.wav") - 12 # -22 LUFS

# Render multi-stem master
mix = music.overlay(sfx).overlay(voice)
mix.export("final_soundtrack.wav", format="wav")
```

Colab Notebook: `notebooks/L06_Speech_and_Audio_AI_Pipeline.ipynb` runs for $0 on free GPUs, includes synthetic stem generators, and builds final videos.

Audio Quality Assurance Checklist: 4 Critical Acoustic Defects

An engineering audit checklist before releasing generative audio assets to broadcast.

1. DIGITAL CLIPPING

Peaks > -1.0 dBFS

Symptom: Harsh crackling on mobile speakers. **Remedy:** Brickwall limiter on master bus set to -1.0 dBFS.

2. PHASE COMB-FILTERING

Metallic 3kHz Resonance

Symptom: Hollow telephone-like timbre. **Remedy:** Upgrade vocoder to 48kHz and notch offending frequencies.

3. PROSODY DRIFT

Erratic Pitch Shifts

Symptom: Narrator abruptly jumps from calm to shouting. **Remedy:** Increase Stability slider to 0.45–0.50.

4. LIP-SYNC LATENCY

Visual Lag > 40ms

Symptom: Mouth closes before acoustic consonants land. **Remedy:** Nudge track against Whisper timestamps.

PRACTICAL EXERCISE • HOMEWORK ASSIGNMENT #6

Homework Assignment #6: Producing a 3-Stem Soundtrack for Lesson 05 Video

Take your 5-second video render from Lesson 05 and engineer a professional 3-layer audio package.

SUBMISSION REQUIREMENTS

3 Isolated Stems + Muxed Video

1. **Stem 1 (Voiceover):** One concise narrative line in ElevenLabs / Cartesia / Kokoro.
2. **Stem 2 (Foley SFX):** Physical sound effect via 3-slot formula in ElevenLabs SFX.
3. **Stem 3 (Music):** Instrumental score in Suno v4 / Udio ducked under voice.
4. Render final video container: `final_video_with_audio.mp4`.

GITHUB REPOSITORY FOLDER STRUCTURE

```
genai-homeworks/L06/
├── 01_voiceover.mp3          # Voiceover
├── 02_foley_sfx.mp3          # Foley SFX
├── 03_music_bg.mp3           # Music
├── final_video_with_audio.mp4# Final Render
├── audio_spec.md             # Specification
└── audio_qa.md               # QA Audit
```

Submission: Add `ihar_rubanovich@epam.com` as a collaborator on your GitHub repository. Feedback within 48 hours.

MODULE 2 CONCLUSION • TRANSITION TO SOFTWARE ENGINEERING

Module 2 Summary: 4 Engineering Laws of Audio and Lesson 07 Preview

Module 2 (Multimodality) is complete! You can now produce studio-grade images, video, and broadcast audio.

4 ENGINEERING LAWS OF AUDIO IN GENAI

Core Principles to Remember

1. **VAD First:** Never pass raw audio into Whisper without upstream Silero VAD silence pruning.
2. **Punctuation Governs Breath:** Shape TTS rhythm and cadence using ellipses and em-dashes.
3. **Instrumental Only for Video:** Never generate AI vocals in background music beneath dialogue.
4. **Sidechain Ducking:** Always attenuate background music by -5 dB whenever speech occurs.

PREVIEW: MODULE 3 • LESSON 07 COPILOT ECOSYSTEM

AI in Software Engineering

- **Microsoft 365 Copilot & GitHub Copilot:** Semantic Index and Graph grounding.
- **Agent Mode in VS Code:** Autonomous bug fixes and terminal agent workflows.
- **Real-World Win:** EPAM CodeMie @ Dawn Foods (35% SDLC acceleration).